

JOONWON LEE

Piscataway, NJ 08854 | 612-438-9144 | joonwon.lee22@gmail.com | github.com/cl20813

SUMMARY

Statistics Ph.D. candidate and data scientist specializing in rigorous statistical testing, causal and observational analysis, uncertainty quantification, and diagnostics for noisy, incomplete data. Develops Python/PyTorch analytical systems that translate ambiguous risk questions into measurable hypotheses, quantify false-positive/false-negative tradeoffs, and distinguish systematic data artifacts from model failures. Experience communicating model evidence and limitations in financial-risk and insurance settings. Available to begin full-time employment in February 2027.

EDUCATION

Ph.D. in Statistics, Rutgers University, Piscataway, NJ

Expected May 2027 | GPA: 3.8/4.0

M.S. in Statistics, University of Minnesota, Minneapolis, MN

Sep 2021 | GPA: 3.7/4.0

TECHNICAL SKILLS

Programming and Data: Python (NumPy, Pandas, SciPy, PyTorch, LightGBM, scikit-learn), SQL, C++/pybind11 integration, Linux, Git, AWS EC2, HPC/SLURM; data cleaning, missing-data workflows, nearest-neighbor matching, and large-scale pipelines

Statistical and Causal Methods: Sequential hypothesis testing, Type I/II error analysis, uncertainty quantification, Monte Carlo simulation, fixed-effects regression, propensity-score matching, GLM/GBM, machine learning, model validation, diagnostics, and performance evaluation

INDUSTRY EXPERIENCE

JPMorgan Chase - Summer Quantitative Analytics Associate, Model Risk Governance and Review

Summer 2026

- Developed and validated a Sequential Probability Ratio Test for binary risk indicators, translating an ambiguous monitoring problem into explicit hypotheses, likelihood-ratio boundaries, and controlled decisions.
- Combined Brownian-motion approximation, dynamic-programming probability propagation, and simulation to quantify boundary-crossing and stopping-time uncertainty, Type I/II error, and early-decision tradeoffs.

Travelers - Data Science Leadership Program, Business Insurance Pricing Team

Summer 2025

- Developed an end-to-end LightGBM pipeline across 2.46M+ records, including data preparation, feature processing, GLM benchmarking, model selection, validation, and performance diagnostics on AWS EC2.
- Translated performance diagnostics and feature-attribution findings into pricing and risk-segmentation recommendations for business stakeholders, including clear communication of model limitations.

STATISTICAL RESEARCH AND APPLIED ANALYSIS

Longitudinal Observational Study - EITC and Household Labor Supply

2021

- Estimated the relationship between Earned Income Tax Credit exposure and household labor supply using fixed-effects regression and propensity-score matching with household income and demographic covariates; defined treatment and comparison groups and interpreted results under observational-study assumptions.

Ph.D. Dissertation Research - Data Quality, Root-Cause Diagnostics, and Scalable Modeling

Sep 2024 - Present

- **Messy-Data Pipeline:** Built observation masks, quality filters, spatial-distance thresholds, hour-specific longitude offsets, and nearest-neighbor center matching to map incomplete satellite observations onto a regular grid.
- **Root-Cause Diagnostics:** Distinguished random measurement noise from systematic acquisition artifacts, including scan-geometry trends with distance from instrument nadir and hour-specific coordinate shifts, before attributing remaining discrepancies to statistical-model misspecification.
- **Scalable Modeling:** Developed an advection-aware Vecchia likelihood approximation for nonseparable Gaussian processes, using ordered conditional models to fit up to 145,008 observations per day in reusable Python/PyTorch CPU/GPU and HPC workflows.

ADDITIONAL

Coursework: Machine Learning, Probability Theory, Stochastic Processes, Statistical Computing, Advanced Theory of Statistics I-II, Data Structures and Algorithms, Econometrics

Languages: Korean (Native); English (Fluent); Chinese (Basic)